Reasoning via Implicit Attention Logic

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Abstract

An implicit deep learning model defines an attention mechanism for a LLM. In turn, the LLM is a semantics for a dependent type theory. This yields a novel reasoning model defined implicitly.

Part 1
Large Language Model - Adam

Notation:
Suppose a corpus is chunked into data sets each of size N,
let rxN input token matrix U = [x_1, ... , x_N],
fixed point matrix K = [k_1, ... , k_N], and
for n > r, nxN output matrix V = [v_1, ... , v_N].

Implicit deep learning model [1]
For n ε [N],
k_n = ϕ (Ak_n + Bx_n)
v_n = Ck_n + Dx_n

In matrix form (1):
K = ϕ(AK + BU)
V = CK + DU

with parameterization A := T - diag(|T| 1n) + γ Idn,
o < γ < 1, to make A well-posed for ϕ,
that is fixed point matrix K exists and is unique [2].

Attention mechanism
For mini-batch SGD, suppose N is partitioned into batches of size 1 < B < N.

For i in [B], define q_i := Bx_i and q′_i := Dx_i, then
k_i= ϕ (Ak_i + q_i)
v_i = Ck_i + q′_i

For i in [B], define probabilities πij = softmax over j ε [B] score(q′_i, k_j),
a distribution of the i-th updated query q′_i over keys k_j, j ε [B].

Define h_i, contextual representation of token x_i, as:
h_i := Σjε[B] πij v_j, where v_j = Ck_j + q′_j,
a probability weighted sum of values.

Let θ = { (W1,b1), (W2, b2) } denote parameters
for a feedforward network defined by

FFNθ(h) := ReLu(hW1 + b1)W2 + b2

Attention layer:
Input h_i
For i ε [B], define probabilities αi,j = softmax over j ε [B] score(h_i, k_j),
where score is a similarity function.
Let h^i = Σ j ε [B] αi,j v_j and
define update h′i = FFNθ(h^i).
Output h′_i

Stack L attention layers with parameters Θ = {θ1, ... , θL}, composing updates FFNθ and now let z_i = h′_i denote the final L-th update.
After L attention layers, for i ε [B] define probabilities
βi,j = softmax over j ε [B] score(z_i, k_j) and
define j*i := arg max over j ε [B] βi,j.
Define token continuation o_i+1 := xj*i.
The LLM Adam maps x_i -> h_i -> z_i -> o_i+i

Define average Renyi α-entropy of {x_i} by:
E({x_}) = (1/B) Σ i ε [B] Hi where
Hi := Hα ({βi,j}) and Hα is Renyi α-entropy.

Adam training problem:
Let λ > 0, learn matrices A, B, C, D, fixed points kj,
and feedforward layer parameters Θ to minimize:

Σi ε [B] {1 - Cos(v_i, Emb(x_i))} +
Perplexity(o2, ... , oB+1) + λE({x_i})

where Emb is a word embedding, like word2vec.
Value v_i simulates word embedding Emb(x_i).
The objective is to minimize the perplexity of attention layer outputs regularized by average Renyi α-entropy.

Summary:
In LLM Adam, an input token x_i gets mapped to update z_i, after L attention layers. Our distribution of interest, that links attention and logic, will be β the distribution of updated queries over keys, but not in its present form, we need an implicit recurrent deep learning proxy to carry out entailment and abduction, a good segue to our logic.

Part 2
Large Reasoning Model - Eve

Open Horn Type Theory (OHTT)

Poernomo, 2025 [3]

Let J range over judgments. J may be any judgment form in any formal system. In a type-theoretic setting, these include formation judgments (A type), typing judgments (t : A), and equality judgments(s ≡ t : A). Let Γ be a context — a finite, consistent collection of assumptions and prior witnesses.

There are two primitive judgment forms:

$\Gamma \vdash^{+} J$ and $\Gamma \vdash^{-} J$

The first is read: "J is witnessed coherent in context Γ." The second: "J is witnessed gapped in context Γ."

Both forms are proof-relevant. If $\gamma : \Gamma \vdash^{+} J$ and $\gamma' : \Gamma \vdash^{+} J$, we do not assume $\gamma = \gamma'$. Different witnesses to the same coherence may carry different structure. The same holds for gap witnesses: ω and ω′ witnessing the same gap may differ.

A judgment J is said to be open in context Γ if Γ contains neither a coherence witness for J nor a gap witness for J.

Exclusion Axiom: Coherence and gap, for the same judgment in the same context, are mutually exclusive.

Let ⇒ denote a compositional relation between judgments. A semantic instantiation of OHTT is entailment: J ⇒ K if the meaning of J implies the
meaning of K.
The specific choice of ⇒ depends on the semantic domain application, for example word-tokens.

For judgments J,K,L and compositional relation ⇒, the open horn type is:

$\Lambda(J,K,L) := (\Gamma \vdash^{+} J \Rightarrow K) \times (\Gamma \vdash^{+} K \Rightarrow L) \times (\Gamma \vdash^{-} J \Rightarrow L)$

An inhabitant of Λ(J,K,L) is a triple (γ1,γ2,ω) where γ1 witnesses that J composes to K, γ2 witnesses that K composes to L, and ω witnesses that the direct composition from J to L is gapped.

A formal semantic interpretation of OHTT [3]:
Definition (Interpretation).
An OHTT context Γ is interpreted as a ruptured Kan complex Γ .
A type A in context Γ is interpreted as a ruptured fibration pA : Γ.A→Γ .
A term t : A in context Γ is interpreted as a coherent section: a map st : Γ→Γ.A with pA∘st =id and st landing in Coh.

Theorem (Soundness).
If Γ ⊢+ J is derivable in OHTT, then J holds in the interpretation.
If Γ ⊢– J is derivable in OHTT, then J is gapped in the interpretation.
Coherence and gap cannot both be derivable (Exclusion).
Proof sketch. Induction on derivations. Each OHTT rule corresponds to a construction on ruptured fibrations that preserves coherence or gap as appropriate. The Exclusion Law corresponds to the exclusion condition in ruptured simplicial sets.

Let A be a space (Kan complex) modeling a semantic domain—say, the space of word-tokens.
We now introduce Eve a semiotic reasoning model on top of LLM Adam but first:

Trichotomy mnemonic*

Coherent: P *entails* Q iff 'Q is a subset of P'

Gapped: P *contradicts* Q iff 'P Λ Q and Q - P are non-empty'

Open: P and Q are *neutral* iff 'P Λ Q is empty'

*We will explain the ' ... ' shorthand in the following Eve LLM semantics.

LLM Semantics
We first define **Eve**, an implicitRNN [1] to proxy
LLM Adam contextual representations z, as follows (2):
For i ε [N], tokens x_i ε R^p.
Initial condition: h_0 := 0 ε R^q
Input: u_i := [x_i h_i-1]'
Equilibrium equation:
s_i = ϕ(A s_i + B u_i)
Prediction equation:

h_i = S(h_i-1, x_i) = C s_i + D u_i
(with parameterization [2] A := Γ- diag(|Γ| 1)+ γ Id, where
0 < γ < 1, ensures A well-posed for nonlinear activation ϕ, that is, the fixed point equation has a solution and it is unique).

Eve training problem:
Let λ > 0, learn matrices A, B, C, D, and fixed points s_j, to minimize:

Σ i ε [N] (1 - Cos(h_i, z_i)) (*)

where Cos(a,b) = <a, b>/(||a|| ||b||),
to make h_i = Eve(x_i) a recurrent proxy of Adam update z_i.

Entailment [4]:
Suppose Eve recursively processes two token string sentences,
premise P = (p_1, ... , p_r) and hypothesis H = (h_1, ... , h_s) as follows:
h_P,j = S(h_P,j-1, p_j), j ε [r] and
h_Q ,j = S(h_Q,j-1, q_j), j ε [s],
to obtain h_P := h_P,r and h_Q := h_Q,s.

Define probabilities β(h) = softmax over j score(h, k_j),
β(h) = (β_1(h), ... , β_m(h)) is the distribution
of Eve proxy h for Adam z, over keys k_j .
Fix 0 < τ < 1. Define F_τ(h) := { k_j | β_j(h) > τ }.
Using a decision tree, we now explain the trichotomy mnemonic shorthand.

Define **entailment** denoted P =>_τ Q iff
F_τ(h_Q) is a subset of F_τ(h_P).
We will drop the subscript τ unless necessary.

Define **contradiction** denoted P => not Q iff
F_τ(h_Q) is a not subset of F_τ(h_P) and
F(h_P) Λ F(h_Q) is non-empty.

P and Q are said to be **neutral** iff
intersection F(h_P) Λ F(h_Q) is empty.

On training triples (P, Q, y), for example from the
Stanford Natural Language Inference (SNLI) Corpus,

where y ε { entail, neutral, contradiction}
learn **Entailment** defined by:

Entailment(h_P, h_Q) :=
entail, if P => Q,
contradiction, if P => not Q,
neutral otherwise,

to minimize (*) and Entailment classification loss,
(for example, # entails misclassified as a contradiction +
contradictions misclassified as an entail +
neutrals misclassified as entail or contradiction).
Thereby, Eve learns textual entailment.

Theory OHTT has a simplicial model semantics, Kan complexes of word-tokens are a domain application. The OHTT compositional relation ⇒ can then be interpreted as entailment in Eve, enabling logical reasoning.

Suppose in Eve, P =>_τ Q and Q =>_μ R, what is the semantics of open horn type
Λ(P, Q, R) := (Γ ⊢+ P ⇒ Q)×(Γ ⊢+ Q ⇒ R)×(Γ ⊢– P ⇒ R)?

i) if μ = τ then P =>_τ R, the horn is coherently filled,
ii) if μ > τ then writing < for 'is a subset of'
F_μ(h_R) < F_τ(h_R) and
F_μ(h_R) < F_μ(h_Q) < F_τ(h_Q) < F_τ(h_P) hence F_μ(h_R) < F_τ(h_P) Λ F_τ(h_R),
the horn is gap witnessed,
iii) otherwise μ < τ, neither coherence nor gap is witnessed for P ⇒ R, and the horn is open.

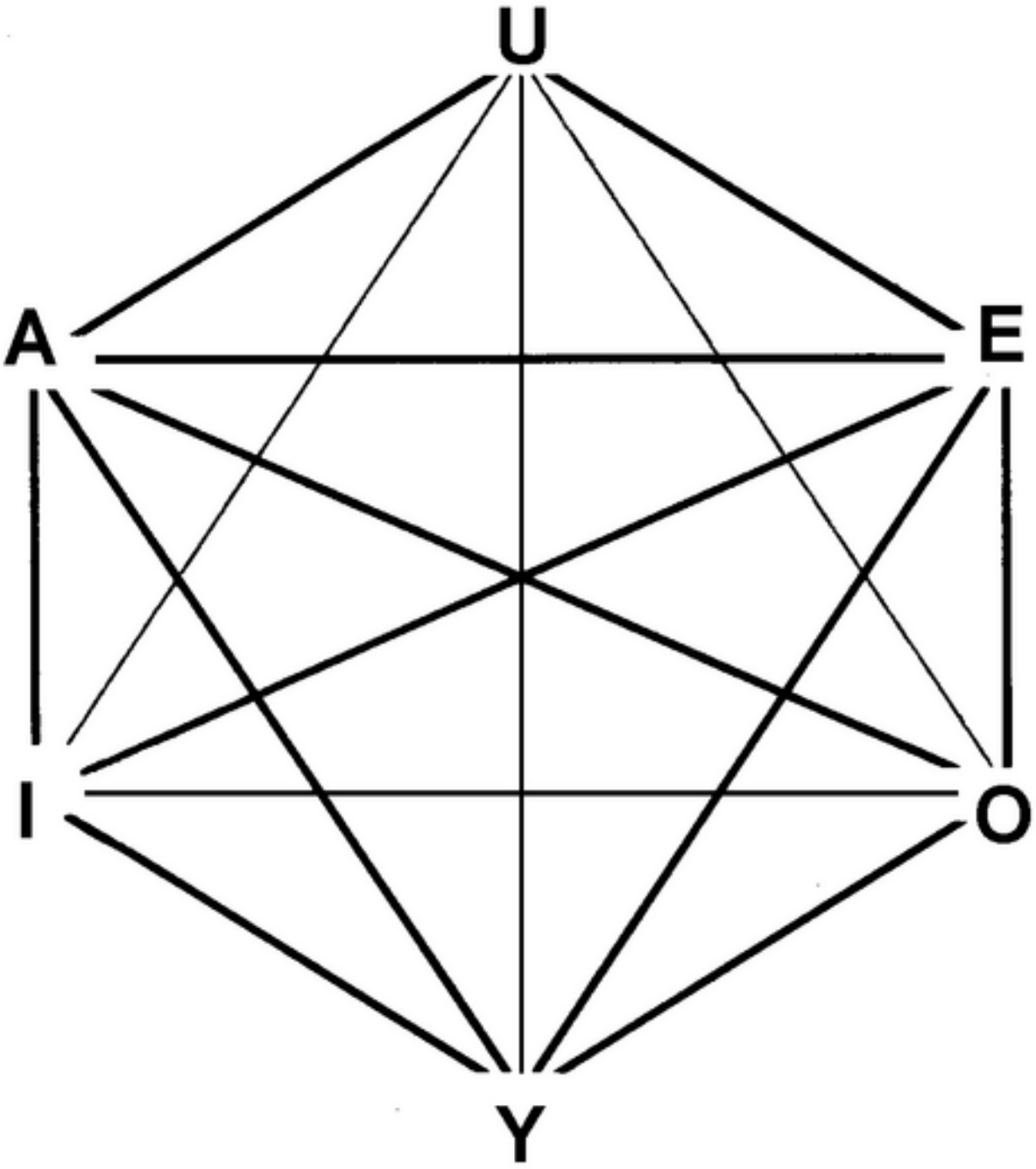
Curiously, attention and logic are entwined by β, a function of the similarity function chosen.
(Fibrations are both ubiquitous and foundational.)

Abductive inference [4]:
Given a hypothesis Q, we wish to find a premise P such that P => Q. Token string Q may be auto-regressively continued by Eve to generate Q, P1, P2, ... , Pn, where n > 1, and Pj denotes a continuation sentence. Choose 0 < τ < 1, for each j ε [n],
check if Pj => Q. Suppose Ω = { Pj | Pj => Q } is non-empty.
Let m := min over P ε Ω, | F(h_P) | and Pj* be the first Pj In Ω with | F(h_Pj) | = m.

Abduction: Given a hypothesis Q, we want to determine a premise P that entails Q. If Ω is non-empty, choose premise Pj* as above that entails Q. Pj* => Q is a parsimonious selection of a premise to explain Q.

So Eve has textual entailment and abductive reasoning.

Semiotic Logical Hexagon [5] for OHTT



A := Coherence witness $\gamma : \Gamma \vdash + P \Rightarrow Q$

E := Gap witness $\omega : \Gamma \vdash_{-} P \Rightarrow Q$

I = ¬ E and O = ¬A

U = A ⋁ E - Determinacy, either coherent or gapped or both

Y = I ⋀ O - Open, neither coherence nor gap is witnessed for P ⇒ Q

Semiotics [6]
Suppose random variable X has distribution p_X,
denoted $X \sim p_X$.
Let H_α denote Renyi α-entropy.
For $(X, Y) \sim p_{X,Y} = p_X p_{Y|X}$, Arimoto mutual information
$I_{A,\alpha}(X; Y) := H_\alpha(X) - H_{A,\alpha}(X|Y)$
where $H_{A,\alpha}$ is Arimoto conditional α-entropy.

To quantify the uncertainty of text, define semiotic breadth
$S(Q) := H_\alpha(h_Q)$ where $h_Q \sim \beta(h_Q)$.
We say P is an 'interpretation' [2] of Q if P => Q
and define semiotic decipherability $D(P) := I_{A,\alpha}(h_Q; h_P)$
to quantify the reduction in uncertainty about an interpretation
that results from observing a message.
To profile semiotic channel capacity $C_{A,\alpha} :=$ max over h_Q, $I_{A,\alpha}(h_Q; h_P)$
map S against D, creating a LLM semiotic cloud signature.

LLM Adam to LRM Eve [cf 8]
An implicit deep learning model (1), defined our attention mechanism keys and values.
Implicit recurrent model (2) defined Eve state h to proxy LLM Adam attention update z.
Then the distribution β of h over keys k, connected (implicit) attention to (OHTT) logic.

In summary, LRM Eve on top of LLM Adam is able to reason via Open Horn Type Theory and infer abductively via textual entailment; a computational epistemology independent of mind or intent. Minds express meaning in models of theories. Sound interpretations enable syntactic reasoning about model semantics. Entailment enables abductive reasoning and the semiotic logical hexagon enables counterfactual reasoning, all by machine. Eve pointing to the moon. Sometimes we need a quick LLM answer, sometimes we need a careful LRM reason,

to wit Kahneman's thinking, fast and slow! [7]

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